Experiment Results: Concordia Agents in the Repeated Prisoner's Dilemma

February 2026 · Status: Post-analysis (v2: pre/post comparison)

Research Question

How do LLM-based agents with associative memory and goal-oriented reasoning behave in a finitely repeated Prisoner's Dilemma? Standard game theory predicts mutual defection via backward induction in a finitely repeated PD, yet behavioral experiments with humans routinely find substantial cooperation, particularly in early and middle rounds, with unraveling toward defection near the end ("endgame effect"). This experiment tests whether Concordia's `RationalEntity` agents—which combine situation perception, memory retrieval, option evaluation, and goal-directed reasoning—produce human-like cooperation patterns or converge to the Nash equilibrium.

Secondary question (v2): Does the implementation of "simultaneous" moves matter? The initial run ("pre-fix") used a fixed act order (A always first) and shared observation labels ("Player A chose X, Player B chose Y"). The post-fix run randomizes act order each round and uses role-relative observations ("You chose X. The other player chose Y."). We compare the two conditions to determine whether the positional asymmetry found in v1 was an implementation artifact.

Experimental Design

Game: Repeated Prisoner's Dilemma, 10 rounds, 2 symmetric players (Player A and Player B).

Actions: Each round, both players simultaneously choose COOPERATE or DEFECT.

Payoff matrix (per round):

	B Cooperates	B Defects
A Cooperates	\$3 / \$3	\$0 / \$5
A Defects	\$5 / \$0	\$1 / \$1

Maximum possible per-player earnings: \$30 (mutual cooperation all 10 rounds). Mutual defection benchmark: \$10.

Agent architecture: Concordia `RationalEntity` prefab with components: Instructions → Observation → SituationPerception → RelevantMemories → AvailableOptionsPerception → BestOptionPerception → ConcatAct. Each agent's goal string specifies the payoff matrix and earnings-maximization objective with no prescribed strategy.

Model: `gpt-4o-mini`. N = 30 sessions per condition.

Conditions:

1. **Pre-fix** (seed 42): Fixed act order (A always first), shared observation labels referencing "Player A" and "Player B" by name.
2. **Post-fix** (seed 43): Randomized act order each round (`random.choice([True, False])`), role-relative observations ("You chose X. The other player chose Y.").

Data Overview

Sample: 60 simulations (30 pre-fix + 30 post-fix), 600 total rounds. No sessions excluded—all completed 10 rounds with valid outputs. Zero invalid actions across all 600 rounds.

Summary statistics by condition:

	Pre-fix		Post-fix	
Metric	Mean	SD	Mean	SD
A earnings ($)	22.3	9.7	22.9	7.3
B earnings ($)	26.3	4.6	22.6	7.5
Joint earnings ($)	48.6	13.8	45.5	12.7
Efficiency	0.81	0.23	0.76	0.21
A cooperation rate	0.71	0.34	0.59	0.37
B cooperation rate	0.63	0.44	0.60	0.37
Mutual coop rounds	5.9	4.8	5.0	4.5
Full coop games	16/30 (53%)		6/30 (20%)	
A exploits B	0.4	1.0	0.9	1.3
B exploits A	1.2	1.6	0.9	1.2

Results

Primary Outcome: Asymmetry Eliminated

The pre-fix condition showed a significant B > A earnings asymmetry: Player B earned $4.0 more on average (paired $t(29) = -3.45$, $p = 0.002$). The post-fix condition *eliminated* this asymmetry entirely: the A−B gap was +$0.3 ($t(29) = 0.25$, $p = 0.81$).

The between-condition comparison of the A−B earnings gap is significant: $t(58) = -2.43$, $p = 0.018$. This confirms that the asymmetry was a **positional artifact** caused by (a) A always acting first and (b) non-relative observation labels, not by statistical noise or inherent agent asymmetry.

The exploitation rates tell the same story: pre-fix B exploited A 3× more often than A exploited B (1.2 vs. 0.4 rounds/game). Post-fix, exploitation is symmetric (0.9 vs. 0.9).

Secondary Outcome: Cooperation Rate Declined

An unexpected finding: the post-fix condition shows *lower* cooperation and fewer full-cooperation games. Full cooperation dropped from 53% (16/30) to 20% (6/30), a significant difference (Fisher's exact $p = 0.015$).

However, the *average* cooperation rates are not significantly different: A coop rate 0.71 vs. 0.59 ($t = 1.35$, $p = 0.18$), B coop rate 0.63 vs. 0.60 ($t = 0.35$, $p = 0.73$). The difference is driven by the tail of the distribution: the pre-fix condition had a large cluster of perfect-cooperation games that partially disappeared post-fix.

Mean efficiency also declined from 81% to 76%, but this is not significant ($t = 0.89$, $p = 0.37$).

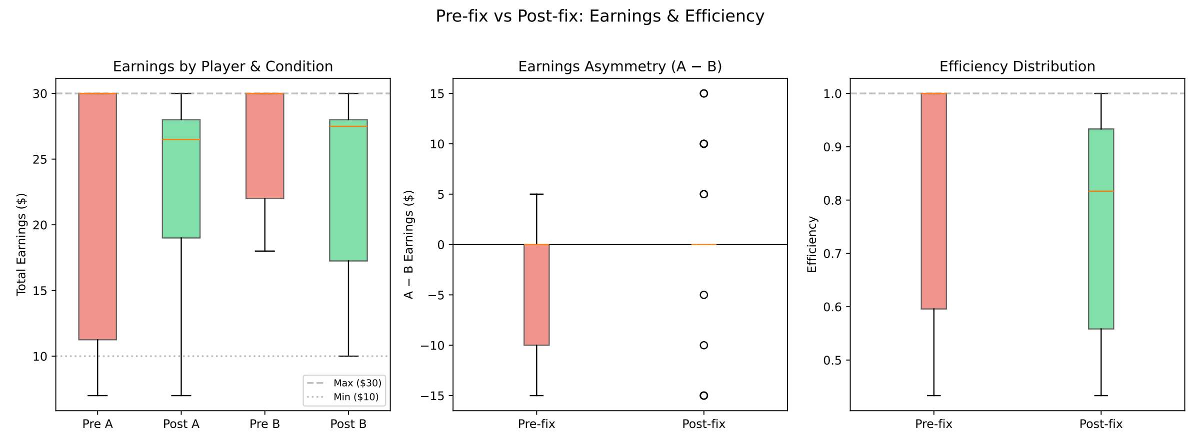

Figure 1: Left: Earnings by player and condition (pre-fix B > A gap visible; post-fix symmetric). Center: A−B earnings gap (pre-fix significantly negative; post-fix centered on zero). Right: Efficiency (slightly lower post-fix but not significant).

Round-by-Round Dynamics

Round	Pre A	Post A	Pre B	Post B
1	80%	40%	70%	47%
2	67%	50%	57%	57%
3	73%	67%	63%	73%
4	67%	67%	67%	57%
5	70%	60%	60%	63%
6	77%	57%	67%	63%
7	73%	60%	60%	57%
8	73%	67%	63%	60%
9	70%	63%	63%	60%
10	63%	60%	63%	60%

The most striking difference is in **Round 1**: pre-fix A cooperated 80% vs. post-fix A cooperated only 40%. This 40-percentage-point drop in first-round cooperation for A suggests that the pre-fix condition's fixed act-order (A always first) somehow encouraged A to open with cooperation, possibly because A's initial move was "fresher" in the LLM context. Post-fix, with randomized ordering and role-relative framing, first-round cooperation is closer to 50/50 for both players.

Neither condition shows an endgame effect—cooperation rates remain roughly flat from round 2 onward.

Choice Sequence Patterns

The heatmaps confirm the bimodal pattern persists in both conditions but with different proportions. Pre-fix had roughly a 50/50 split between full cooperators and defectors. Post-fix shifted to roughly 20/80, with more mixed-strategy games and fewer lock-in cooperators.

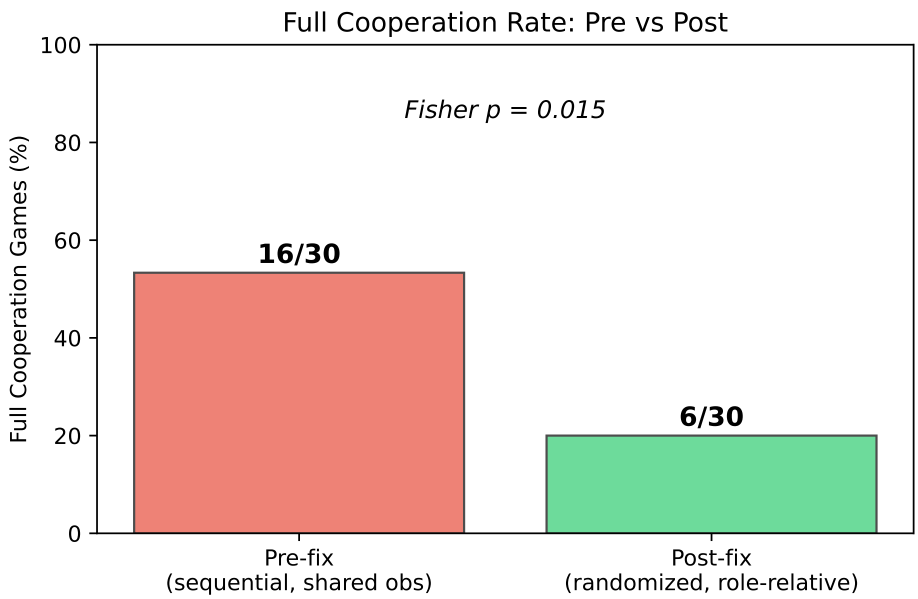

Figure 2: Full cooperation rate by condition. The drop from 53% to 20% is significant (Fisher's exact $p = 0.015$).

Evaluation

Did We Learn Something Interesting?

Yes—two clear findings:

1. **The B > A asymmetry was entirely an implementation artifact.** The combination of fixed act-order and named observation labels created a systematic advantage for Player B. Randomizing act-order and using role-relative observations eliminated the asymmetry completely ($p = 0.018$ for the between-condition difference). **Methodological lesson:** In Concordia simulations of simultaneous-move games, both act-order randomization and role-relative observations are necessary. This should be standard practice.

2. **Implementation details affect cooperation rates substantially.** The post-fix condition dropped full cooperation from 53% to 20% ($p = 0.015$), driven primarily by a collapse in Round 1 cooperation for Player A (80% → 40%). This means the high cooperation rate in the pre-fix condition was *partially inflated* by the positional bias. The "true" cooperation rate under fair conditions is closer to 59–60%, with 20% of games achieving full cooperation—still well above the Nash equilibrium of 0%, but less dramatic than the initial 53% figure.

The corrected picture: Concordia `RationalEntity` agents cooperate at ∼60% in a 10-round PD with no endgame effect. Full cooperation occurs in ∼20% of games. The bimodal pattern (cooperate from start or defect from start) persists. Both findings diverge from Nash predictions and from typical human behavior (where cooperation starts high and decays).

Limitations

- **Confounded comparison:** The pre-fix and post-fix runs changed *two things simultaneously*—act order randomization and role-relative observations. We cannot determine which

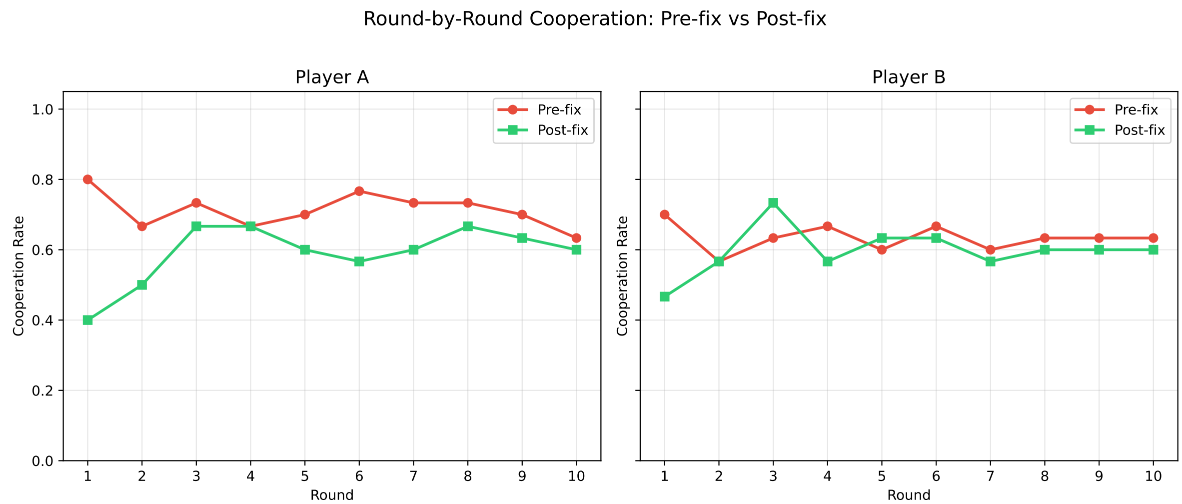

Figure 3: Round-by-round cooperation rates. Pre-fix (red) vs. post-fix (green). The largest gap is in Round 1 for Player A.

change drove the cooperation decline (or whether both contributed). A 2×2 factorial with (act order: fixed vs. random) × (observations: named vs. relative) would disentangle these.

- **Different seeds:** Pre-fix used seed 42, post-fix used seed 43. While 30 simulations per condition provides reasonable power, the different seeds mean some variance is attributable to sampling.
- **Single model (gpt-4o-mini):** Results may not generalize to other models.
- **No human baseline:** We cannot say whether 60% cooperation is "human-like" without running the same game with human subjects.

Next Experiment

Recommended archetype: DEEPEN

The corrected post-fix results provide a clean baseline: ∼60% cooperation, no endgame effect, bimodal pattern, symmetric earnings. The next experiment should test whether a well-studied manipulation from behavioral economics—*costly punishment*—shifts cooperation rates as predicted by human experiments.

Proposed experiment: Costly Punishment

After each round, each player can pay $1 to reduce the other player's earnings by $3 (standard altruistic punishment mechanism from Fehr & Gächter, 2002). The hypothesis is that punishment availability will increase cooperation by deterring defection, converting the bimodal distribution (20% full coop, 80% mixed/defect) into a more cooperative distribution.

Conditions:

1. `pd_concordia` (baseline: this experiment, post-fix)
2. `pd_concordia_punish` (same game + punishment stage after each round)

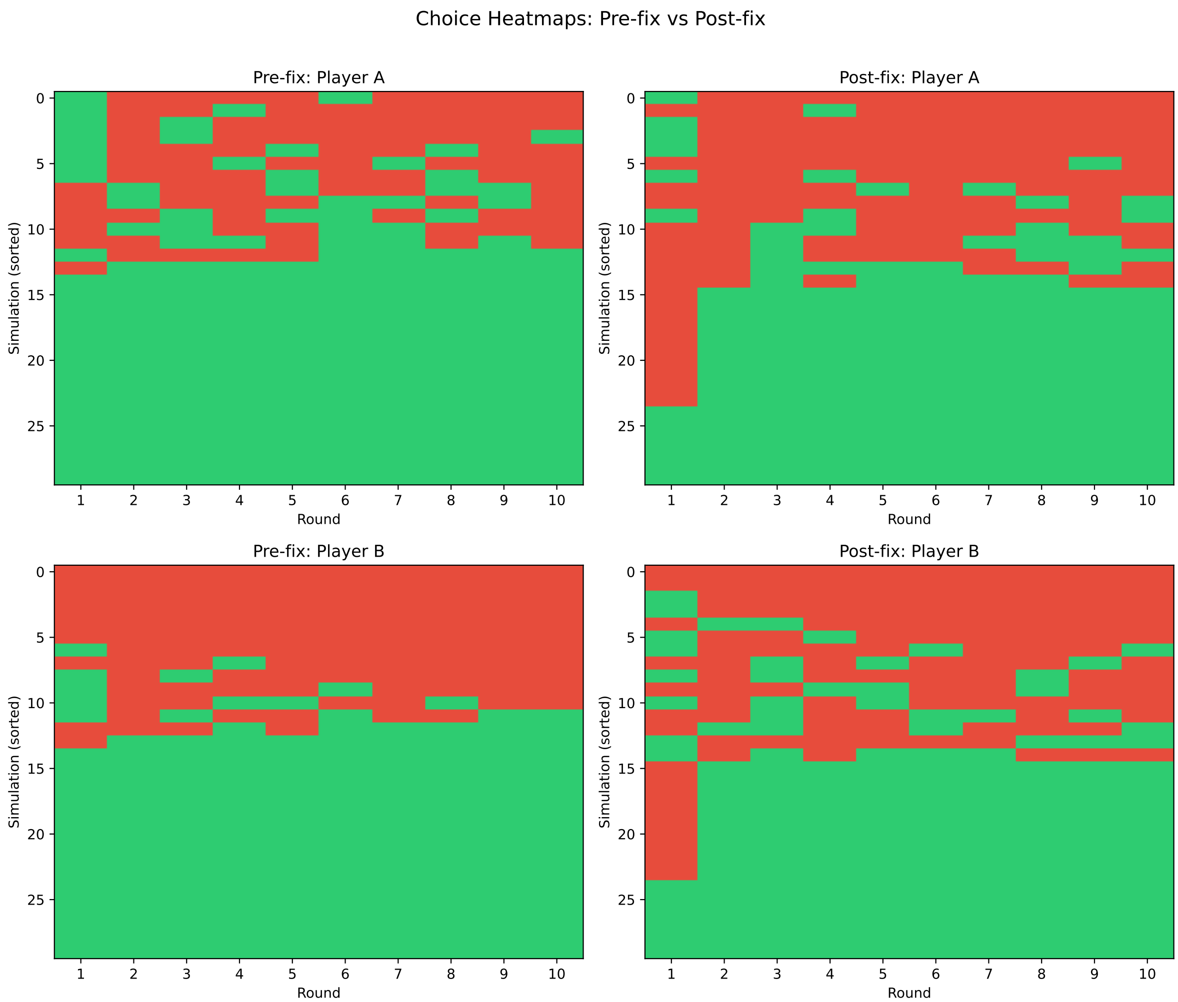

Figure 4: Choice heatmaps (green = cooperate, red = defect). Top row: Player A; bottom row: Player B. Left: pre-fix; right: post-fix. Simulations sorted by total cooperation. The post-fix condition has fewer "solid green" rows (full cooperation games).

Predictions:

1. Full cooperation rate increases from 20% to >50%.
2. Mean cooperation rate increases from 60% to >80%.
3. Punishment is used primarily in early rounds to establish norms, then declines.

Alternative: To cleanly disentangle the act-order vs. observation-format effects, run a 2×2 factorial: (fixed act-order × named observations, fixed × relative, random × named, random × relative). This would confirm which implementation detail drives the cooperation difference.

Next step: Run `/create-concordia-game` for the punishment variant, or run `/hypothesize` with the punishment hypothesis to formalize it first.

Prior Experiment Context (Machine-Readable)

prior_hypothesis: LLM agents with structured cognitive architectures sustain
  non-trivial cooperation in a 10-round PD despite Nash prediction of defection
verdict: YES_INTERESTING
diagnosis: NA
key_finding: Asymmetry was implementation artifact (p=0.018);
  corrected cooperation ~60%, full coop 20%, no endgame effect;
  implementation details (act order, observation format) significantly
  affect cooperation rates (full coop 53% -> 20%, p=0.015)
key_statistic: post_fix_coop_rate=0.60, post_fix_full_coop=6/30=20%,
  asymmetry_eliminated (p=0.81), full_coop_decline (Fisher p=0.015)
dag_variables: X=cognitive_architecture, M=memory_and_reasoning,
  Y=cooperation_rate, Z=payoff_structure, W=implementation_details
testable_implications_results: cooperation>0=CONFIRMED,
  endgame_decline=REFUTED, symmetric_earnings=CONFIRMED_AFTER_FIX,
  implementation_neutral=REFUTED
next_archetype: DEEPEN
proposed_changes: Add costly punishment condition ($1 to punish,
  -$3 to target) to test whether punishment shifts cooperation
  from 20% full-coop to >50%
next_hypothesis_sketch: Costly punishment availability increases
  cooperation rate from 60% to >80% by deterring early defection,
  converting bimodal distribution to unimodal cooperative